HW6 POINT POSITIONING ERROR ANALYSIS

Assigned: 19 October 2026

Due: 2 November 2026

OBJECTIVE

This assignment completes the process of using GPS pseudorange measurements to form a solution. In HW6 we computed a single position solution (also called a point solution) at one specific time (or "epoch"), focusing on implementing all of the necessary pseudorange corrections, constructing the G-matrix (measurement connection matrix or measurement partials), and investigating how well this process converges to the correct answer given an a priori guess that is far from the truth. In this assignment we compute point solutions for a station over the course of an entire day using each of the measurement epochs given in a RINEX observation file. In addition to the mechanics of solving for position we also will look at the resulting position errors to see how they change over the day. To do this, we solve for the position correction (or relative position) with respect to the known receiver position, at each epoch. Corrections or relative positions like this are often expressed in East-North-Up components, which provide more meaningful interpretation of the position errors compared to the ECEF X, Y, Z errors.

We will look at data from NIST and one or two other sites – the locations and observation files are given in attached text files.

DATA

- Date: Wednesday, August 19, 2026 = GPS Week 2432 (or 384), TOW 259200-345600/ DOY 231
- SITES: NIST (in Boulder, CO): ECEF coordinates: [-1288398.567 -4721696.932 4078625.350] (m)]
- (Additional Site to be added shortly)
- RINEX File: NIST00USA_R_20262310000_01D_30S_MO.rnx
- Broadcast ephemeris file: brdc2310.26n (DOY is the 3-digit day of year)

ASSIGNMENT

Models and preparation for solution:

- Use the positions for each location given in the text files as your initial (*a priori*) guess for the station coordinates. You should assume that this position is correct.
- Based on these locations, compute the transformation matrix from ECEF to ENU for each of the two locations. You only need to compute this once – i.e. it doesn't change throughout the day.
- Set up your code to loop through the entire obs file, working with the data from one epoch at a time. (While you are debugging your code, just work with the first few epochs so you don't have to read everything and work with the large amount of data.)
- For each epoch, loop through the satellites computing the following for each one (these are all based on prior homeworks):
    - The satellite position, expected range, satellite clock correction, and relativity correction based on the parameters in the broadcast ephemeris file and the *a priori* receiver location.

  - To correct for the ionospheric delays use the dual frequency ionosphere-free pseudorange combination PIF which is computed using P1 (or C1) and P2.
  - To correct for tropospheric delay corrections use the recommended mapping function and determine an appropriate zenith delay for the location or use the UNB-3 model. Be sure to explain your selections in the code comments and report write up.

- Calculate and store the pre-fit residuals. Remove all satellites below 10 deg elevation and also remove any (ridiculously) large measurement outliers in the pre-fit residuals before computing a least-squares solution. Keep a record of how many measurements are being discarded.

Least Squares Point Solutions:

- At each epoch, use the position given in the class text file as your initial (*a priori*) guess for the station coordinates. This will be used to construct your G-matrix as well as the expected range.
- Compute the least squares position and clock solution: dx, dy, dz, b for each epoch of the day. This solution is an adjustment or correction to the *a priori* guess. Do not iterate this, just use the position given above as the *a priori* for each epoch, and solve for a position & clock correction. We will refer to this position correction as the "relative position" in the following.
- Compute the post-fit residuals: dpr_post = dpr_pre – G [dx dy dz b]'
- Rotate the *relative position* to East-North-Up (ENU also called ENV) coordinates based on transformation matrix you computed above for each location. (Another option is to express the LOS vectors in the G-matrix in ENU, then your relative position and DOPs will both automatically be in ENU coordinates.) You will now have a position correction for each time on the obs file (total of 2880 of them because the measurements are output every 30 seconds).
- Compute the East, North, and Up - **DOP** values for each epoch.
- Compute the standard deviation and RMS value of each of the relative position coordinates and of the pre and post-fit residuals over the entire day.
- See if you can find a good zen_tropo value for each of the sites – explain your method.
- Compare the clock solutions (b) for the two sites over the day.

What to submit: All of your code, well-commented (20pts) and the following plots and results for each location (60pts):

1. Print out your relative position solution in East, North, Up (vertical) coordinates to a precision of 1 cm, the clock bias (b), and the corresponding DOP values (EDOP, NDOP, VDOP, TDOP) **at time 01:00.**
2. Print out the standard deviation and RMS value of each of the relative position coordinates and the pre and post-fit residuals. These are computed over the entire day, so you should have a total of 5 standard deviations and 5 RMS values.
3. Plot the pre-fit residuals as a function of time (hours) and as a function of elevation (deg). (Do not include outliers.)

4. Plot the post-fit residuals as a function of time (hours) and as a function of elevation (deg).
5. Make a scatter plot of all the horizontal relative position solutions (horizontal relative positions, i.e. dNorth vs dEast (North on the y-axis, East on the x-axis)
6. Plot the ENU solutions (m) as a function of time (hours).
   (You can add the expected 2-σ bounds to this plot by multiplying the DOP values by 2 times the standard deviation of the postfit residuals.)
7. Plot the EDOP, NDOP, and VDOP values as a function of time (hours).
8. Plot the number of satellites used in each point solution as a function of time (hours).

9. Plot the clock solutions for both of the sites vs time (hours) on the same graph.
10. Discussion (~1-2 pages, 20 pts) – Describe and explain the results, referring to the figures and tabulated results. First provide a basic overview of the results. Did you observe significant differences in the positioning performance over the course of the day or differences between the sites? What are likely to be the dominant causes of these differences? What similarities and differences do you see between different components of the solutions? Were any of the results surprising? Use what you have learned from previous assignments to back up your analysis.